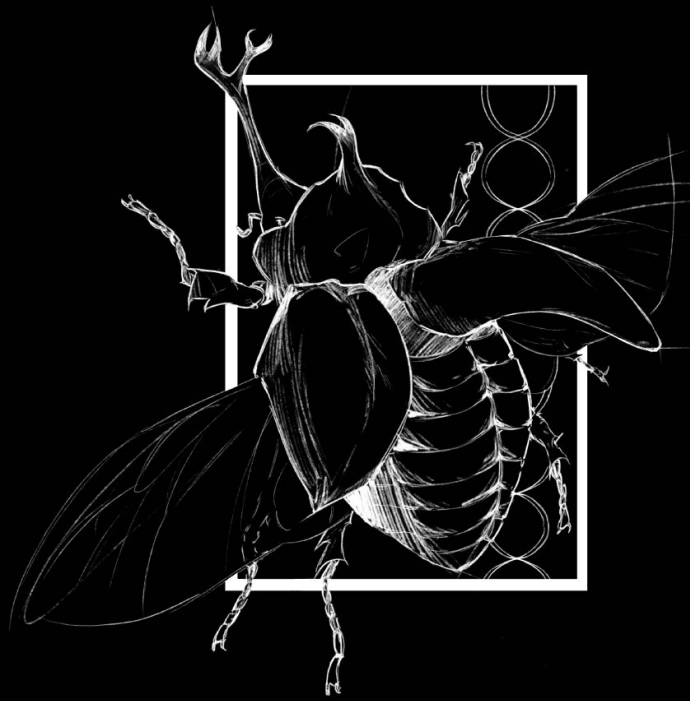




Background

Testudines, or turtles and tortoises, live in a variety of habitats that have influenced their rates of evolution. **This project addresses the question- does habitat type (marine, freshwater, or terrestrial) affect the rate of chromosome number evolution in turtles?**

Current research and hypotheses suggest that species inhabiting larger, more stable environments, such as marine testudines, may experience slower rates of chromosomal evolution compared to those in smaller or more variable habitats like freshwater or terrestrial ecosystems. This hypothesis is known as the Drift-Barrier Hypothesis created by Michael Lynch. Using a comparative method is ideal for testing this question, as it allows us to examine evolutionary patterns across many species while controlling for shared ancestry.



AI Use

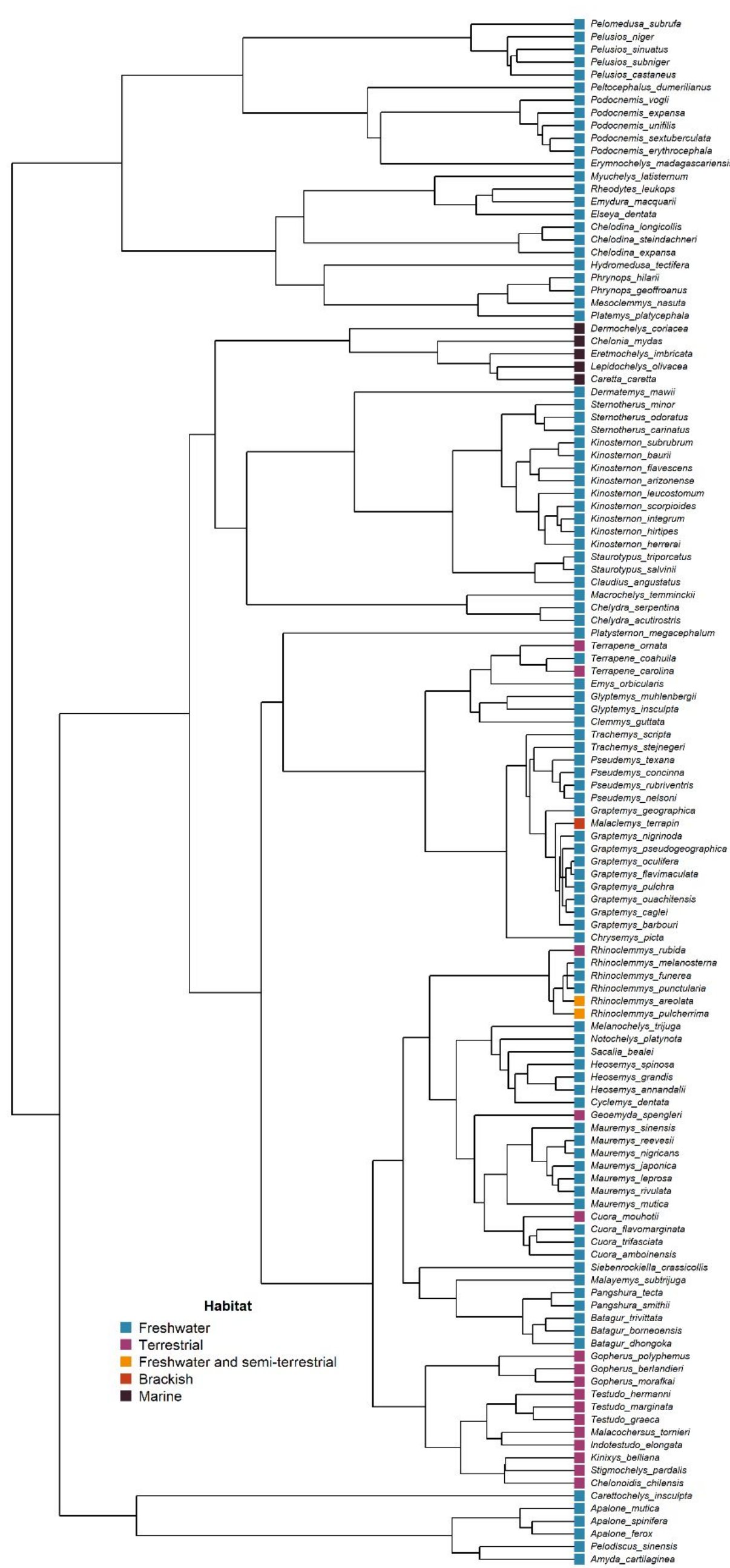
Artificial Intelligence (AI) was used throughout this research. Claude was used to generate code for R Studio to gather data and produce figures. All AI-generated code was reviewed and validated before use to ensure accuracy of the analysis. Additionally, AI was used to assist with interpreting statistical outputs and understanding the biological significance of results.

Study Design & Analysis Plan

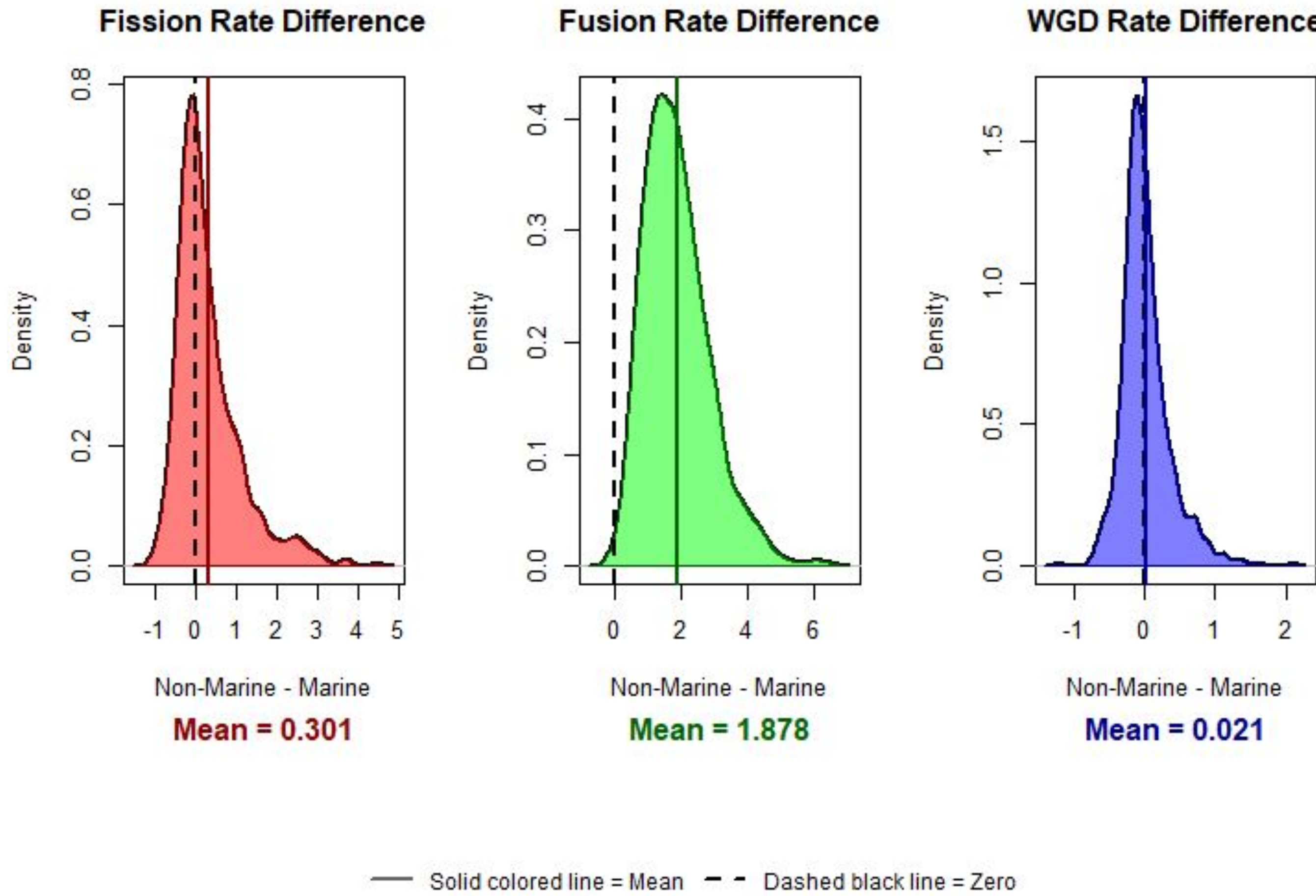
Regarding the study design, the Bayesian Markov Chain Monte Carlo (MCMC) approach implemented in the chromePlus package was utilized to estimate rates of chromosome evolution across testudine species. I compared marine testudines to non-marine testudines (freshwater and terrestrial species) to test whether habitat type influences the rate of chromosomal change. Separate rates were estimated for fission, fusion, and whole genome duplication events. This study tested predictions of the Drift-Barrier Hypothesis, which predicts higher rates of chromosomal change in non-marine species due to smaller, fragmented populations experiencing stronger genetic drift compared to large, connected marine populations. AI-assisted code development facilitated implementation of this phylogenetic comparative framework.



Testudine Phylogeny with Habitat Types



Results



The results reveal notable differences in chromosome evolution rates between marine and non-marine testudines. Most significantly, fusion rates were approximately 6.5 times higher in non-marine species compared to marine species (mean difference = 1.878). Fission rates were slightly elevated in non-marine species (mean difference = 0.301). Contastingly, whole genome duplication (WGD) rates showed no meaningful difference between habitat types (mean difference = 0.021), indicating that this mechanism does not contribute to habitat-specific chromosome evolution.

Discussion

The results demonstrate that non-marine testudines exhibit significantly faster rates of chromosome evolution compared to their marine counterparts, especially in chromosomal fusion. Non-marine species show a 6.5-fold increase in fusion rates, which contributes to a notable reduction in chromosome numbers over time. These elevated fusion rates indicate that non-marine lineages are evolving toward lower karyotype counts, while marine testudines maintain much more stable karyotypes, characterized by balanced rates of chromosome fission and fusion. Importantly, rates of whole genome duplication are similar in both groups, highlighting that the differences in chromosome dynamics are primarily driven by smaller-scale rearrangements, such as fusions and fissions, rather than large-scale genome duplications. When considering all mechanisms together, non-marine testudines undergo chromosomal rearrangements at about 3.2 times the rate observed in marine testudines, resulting in a net pattern of decreased chromosome numbers among non-marine taxa.

These findings are consistent with the Drift-Barrier Hypothesis, which posits that population genetic parameters—mainly effective population size—determine the rate of genomic and chromosomal evolution. Larger populations, typical of marine testudines, are more efficient at purging deleterious mutations and rearrangements due to strong selective barriers, especially given the high connectivity and dispersal found in oceanic habitats. In contrast, populations of non-marine testudines are often smaller, more fragmented, and less connected, which enhances the effects of genetic drift and diminishes the strength of selection. This context permits neutral or even mildly harmful chromosomal rearrangements, such as fusions, to become fixed more readily.

The observation that fusion rates are particularly elevated in non-marine taxa supports theoretical predictions that fusions are more likely to fix under conditions of strong drift, possibly due to their impact on recombination and meiotic processes. Thus, our results provide empirical backing for the Drift-Barrier Hypothesis as an explanation for the divergent patterns of chromosomal evolution seen in testudines. Overall, this research underscores the crucial influence of habitat, population structure, and ecological context in shaping the pace and direction of chromosome evolution, with non-marine testudines undergoing more rapid and dramatic changes in karyotype compared to their marine relatives.

Conclusions

- Non-marine testudines experience much faster chromosome evolution than marine testudines, primarily due to elevated fusion rates driving lower chromosome numbers.
- Marine testudines maintain stable karyotypes, likely because larger, more connected populations strengthen selection against chromosomal rearrangements.
- These patterns support the Drift-Barrier Hypothesis, highlighting the critical role of habitat and population dynamics in shaping chromosome evolution.